# Research question and mixed methods designs

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# Agenda

## Preliminary scouting and RQ

- Background research

## Tentative research design

- Data sites and analyses

# Any questions from the lecture?

# "*Theory and Design*" In Mixed Methods: A Primer for Sociologist.

*Frederiksen, Morten, Jonas Toubøl, and Hjalmar Bang Carlsen.*

## Why Mixed Methods?

- Mixed methods are used to capture the **complexity** of a phenomenon when single methods fall short.
- The choice of methods should be **guided by theory**
- "*A mixed methods design therefore starts with the theoretical part of the study, where the problem is defined*" (p.51)
- Effective mixed methods design begins with a strong theoretical framework that supports the use of multiple data types and analysis methods.

## Types of Research Questions

Having a set of theories is not enough for investigation. To design a study, you must develop research questions that **operationalize** these theories into something that can be empirically illustrated with data.

| | Theory testers | Theory developing |
|---|---|---|
| Descriptive | 1. Is X characterised by Y? | 2. What characterises X?<br>What is X associated with? |
| Explanatory | 3. Does change in X cause change in Y? | 4. What causes Y?<br>How does X cause change in Y? |

**Descriptions** aim to gain more knowledge about a phenomenon.
- E.g.: "*What are the characteristics of poverty in this community*?"

**Explanation** is about understanding how a phenomenon occurs or changes.
- E.g.: "*How does economic policy influence poverty rates*?"

## Design Integration

Mixed methods design is about how sub-studies relate to each other.

- **Complementarity:** illuminate different parts of a phenomenon using different data to analyze different aspects
- **Confirmation:** test if results from different data sources agree - Analyze the same aspect using different data
- **Sequential:** the different parts (or "sub-studies") of your research are done one after the other, not at the same time. The key idea is that the results from the first part influence or inform what happens next.
- **Simultaneous:** sub-studies are conducted at the same time, with results integrated only after completion

## Design Weight

Not all data sources play equal roles in mixed methods research

- **Primary Study**: carries more analytical weight in final conclusions
- **Secondary Study**: provides supporting evidence or context

CHAPTER 3

### Theory and design

This chapter will focus on answering three key questions:

- Why should you choose a mixed methods study?
- What is integration in mixed methods design?
- Which mixed methods design should you choose?

The chapter begins with some general methodological considerations about how we in the social sciences approach our research subjects. Here we further discuss the relationship between theory, evidence and data, which is fundamental to all research, but which becomes particularly central to mixed methods studies. The reason is that in a mixed methods design, it must be very clear how different types of data and analytical methods are connected to the theoretical starting point. This is what we called theoretical integration in chapter 2: that we have a clear theoretical understanding of the phenomenon we are investigating, and that the theory justifies the necessity of collecting different types of data and evidence to shed light on the theoretical ideas about the phenomenon.

It's also the answer to the first question: you should choose to conduct a mixed methods study in order to capture all the aspects of a phenomenon that theory emphasises as important, and not just focus on the parts that one type of data is good at capturing.

A key point in this chapter is therefore that a problem can and should often be operationalised into several research questions to take into account the relevant dimensions of the theoretical basis of the research.

# Background research and RQ

- In your exam groups, spend the next 30 min doing **preliminary scouting research** on your topic, and organizing it as an *early immersion journal* for your netnography.
- Further, for 15 min, formulate a **preliminary research question**, informed by your earlier literature search, and containing a set of **preliminary concepts** and a preliminary idea **motivating** your research.
  - You are allowed to have multiple competing versions of a RQ at this point.

45 min

# Break for 15 minutes

# Potential data sources

1. Based on your RQ, scout for a **datasite** relevant for your topic. Try to find at least 3 different datasites.

2. Describe each datasite and its **relevance** for your RQ.

3. Take each datasite's available data and try to **creatively repurpose** the traces to answer your RQ. This should be at a rather **specific** level, matching information on the datasite to **central concepts** of your RQ.

45 min

# Break for 15 minutes

# Potential designs

1. Based on your previous RQ and datasites, discuss, for 20 min, what would be **potential qualitative and quantitative analyses** that your data would lend itself to, and what is **the reason for mixing** data types and analysis types.

2. In plenum, students report back from the topic stages – do the topics still look viable and interesting, after your research? Pitch your ideas and progress.

3. Keep working on task 1.

45 min

# Tak for i dag!

**Next time**: Find, evaluate, and select digital data sources

**Seminar on Tuesday:** Working on data sites selection

- Kozinets, Chapter 8, "Investigating: Five Steps to Social Media Data Collection." Page 211-242 (31 pages).
- Kozinets´, Chapter 10 "Immersing: Journalizing and Organizing a Reflective Curation". Page 277-306 (29 pages)
- Venturini, T. et al.. 2018. 'A reality check-list for digital methods'. New Media & Society 20(11). https://doi.org/10.1177/1461444818769236. (14 pages)
- Kozinets, Chapter 6, "Ethics: Procedures and Flowcharts, Updates and Rules." Page 163-187 (25 pages)